Kutay Eken

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Professional Summary

Graduate student in Computer Science (M.S. candidate, GPA 3.88) targeting AI/ML Software Engineer roles, with hands-on research and teaching experience in AI, machine learning, and computer vision. Skilled in designing and deploying reinforcement learning agents, probabilistic models, and end-to-end ML pipelines using Python and modern frameworks. Published in international AI conferences and adept at translating theoretical models into real-world software systems.

Technical Skills

Machine Learning & AI: Predictive Modeling, Deep Learning, Reinforcement Learning, NLP, Computer Vision, Probabilistic Modeling
Programming Languages: Python, SQL, Java, C, C++, C#, R, MATLAB
ML Frameworks: PyTorch, TensorFlow, Scikit-learn, Stable Baselines3, Gym
Data & Tools: NumPy, Pandas, OpenCV, Data Pipelines, Git, Google Cloud

Education

University of Utah — Salt Lake City, UT
M.S. in Computer Science, Artificial Intelligence / Deep Learning — *Expected 05/2026*

University of Utah — Salt Lake City, UT
B.S. in Computer Science, Artificial Intelligence & Information Science — *05/2024*

Projects

Deep RL for Autonomous Driving — 01/2025 - 05/2025
Reinforcement Learning
- Developed PPO-based RL agents for autonomous driving in CARLA using continuous control, achieving **0 collisions** and **0 lane departures** in evaluation scenarios.
- Implemented and optimized uncertainty-aware reward functions, executing **500K+ training steps** with Stable Baselines3 to improve safety-driven decision-making in lane-keeping tasks.

UPark — 08/2023 - 05/2024
Computer Vision & Predictive Modeling
- Designed an end-to-end parking analytics system integrating YOLO-based vehicle detection on **1,000+** aerial images and heatmap visualization to analyze usage patterns.
- Trained occupancy forecasting models using TensorFlow and scikit-learn, achieving **85%** prediction accuracy.

Predicting Income Level — 08/2023 - 01/2024
Supervised Learning & Predictive Analytics
- Built and fine-tuned classification models (Logistic Regression, SVM, Random Forest) using scikit-learn and PyTorch, achieving **85+%** accuracy in predicting income from census data.
- Conducted feature engineering and cross-validation with hyperparameter optimization, resulting in improved generalization and a **10% gain** in test performance.

Experience

Teaching Assistant — 08/2023 - Present
Kahlert School of Computing, University of Utah — *Salt Lake City, UT*
- Supported **200+** students in Data Mining, Computer Vision, and OOP courses through labs and office hours, resulting in improved understanding and higher student success rates.
- Designed lab exercises in Python (NumPy, Pandas) and Java OOP, reinforcing applied learning and improving problem-solving skills.

Research Assistant — 01/2022 - 01/2024
Artificial Intelligence Research Lab — *Salt Lake City, UT*
- Developed a Bayesian reasoning agent for Wumpus World, improving safe action selection and reducing environment failures.
- Engineered an autonomous drone system using UWB and computer vision for real-time localization and obstacle avoidance.

Publications

- T.C. Henderson, A. Lessen, I. Rajan, T. Nishida, **K. Eken**. ”Chop-SAT: A New Method for Knowledge-Based Decision Making.” Conference on Intelligent Autonomous Agents, Jul 2023
- T.C. Henderson, T. Nishida, A. Lessen, N. Wernecke, **K. Eken**, D. Sacharny. ”A New Approach to Probabilistic Knowledge-Based Decision Making.” International Conference on Agents and Artificial Intelligence (ICAART), Feb 2023